

1. ~~(2)~~ 5 (1) 2

2. $x^2 - 3x - m(m+3)$

$\rightarrow x^2 + mx - (m+3)x - m(m+3)$

$\rightarrow x^2 + \cancel{mx} - \cancel{mx} - 3x$

$\rightarrow x(x+m) - (m+3)(x+m)$

$\rightarrow (x - (m+3))(x+m)$

Ans - (2) - m, (m+3)

3. For coincident lines, $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$\Rightarrow \frac{2}{(a+b)} = \frac{3}{(a+b-3)} = \frac{7}{4a+b}$

$\rightarrow 2(a+b-3) = 3(a+b)$

$\rightarrow 2a + 2b - 6 = 3a + 3b$

$\rightarrow a + b = -6$

$\frac{3}{(a+b-3)} = \frac{7}{(4a+b)}$

$\Rightarrow 3(4a+b) = 7(a+b-3)$

$\rightarrow 12a + 3b = 7a + 7b - 21$

$\rightarrow 5a - 4b = -21$

$\frac{2}{(a+b)} = \frac{7}{(4a+b)}$

$\rightarrow 2(4a+b) = 7(a+b)$

$\rightarrow 8a + 2b = 7a + 7b$

$\rightarrow a - 5b = 0$

study time

Subject

$$x^4 + 4x^2 + 4x^3$$

Date: / /

MON TUE WED THUR FRI SAT SUN

Ans- (3) $a - 5b = 0$

4. (3) $(\sqrt{2}x + \sqrt{3})^2 + x^2 = 3x^2 - 5x$

5. let the ratio be $k:1$.

Now required coordinate = $P(x, 0)$

$\Rightarrow (x, 0) = \left(\frac{-3k+2}{k+1}, \frac{-3+6k}{k+1} \right)$

$(x, 0) = \left(\frac{2+5k}{k+1}, \frac{-3+6k}{k+1} \right)$

Upon comparing,

$0 = \frac{-3+6k}{k+1}$

$\Rightarrow -3+6k = 0$

$\Rightarrow 6k = 3$

$\Rightarrow k = \frac{1}{2}$

Ratio = $k:1 = \frac{1}{2} = 1:2$

Ans- (1) $1:2$

6. let the coordinates be $A(2, 4)$ and $B(-3, 1)$.

By distance formula, $\sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$

$AB = \sqrt{(2+3)^2 + (4-1)^2}$
 $= \sqrt{25+9} = \sqrt{34}$

Radius = $\frac{\text{diameter}}{2} = \frac{\sqrt{34}}{2}$

Ans- (1) $\frac{\sqrt{34}}{2}$ units

7. By BPT, $\frac{AD}{AB} = \frac{DE}{BC}$

$$\Rightarrow \frac{3}{8} = \frac{DE}{12.8}$$

$$\Rightarrow \frac{3 \times 12.8}{8} = DE$$

Ans- (1) 4.8 cm

8. $\frac{AB}{DE} = \frac{AC}{DF}$

$$\Rightarrow \frac{5}{7.5} = \frac{8}{DF}$$

$$\Rightarrow DF = \frac{8 \times 7.5}{5} = 12 \text{ cm}$$

Ans- (2) DF = 12 cm, $\angle E = 100^\circ$

9. $\sin 30^\circ = \frac{P}{H} = \text{radius}$

$$\tan 30^\circ = \frac{P}{B} = \frac{\text{radius}}{\text{tangent}}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{3}{\text{tangent}}$$

$$\Rightarrow \text{tangent} = 3\sqrt{3} \text{ cm}$$

Ans- (4) $3\sqrt{3} \text{ cm}$

10. (2) $\frac{BE}{EC}$

11. (2) 40°

12. $\tan \theta = 3$

$$\Rightarrow \tan \theta = \frac{3}{4} = \frac{P}{B}$$

Let $P = 3K$ & $B = 4K$

$H^2 = P^2 + B^2$ (by PyT)

$\Rightarrow H^2 = 9K^2 + 16K^2$

$\Rightarrow H = \sqrt{25K^2} = 5K$

$\sin \theta = \frac{P}{H} = \frac{3K}{5K} = \frac{3}{5}$

$\cos \theta = \frac{B}{H} = \frac{4K}{5K} = \frac{4}{5}$

$$\frac{4 \times \frac{3}{5} - \frac{4}{5}}{4 \times \frac{3}{5} + \frac{4}{5}} = \frac{\frac{8}{5} - \frac{4}{5}}{\frac{8}{5} + \frac{4}{5}} = \frac{\frac{4}{5}}{\frac{12}{5}} = \frac{1}{3}$$

Ans- (3) $\frac{1}{3}$

13. Area of the yard not lit = (area of the yard) - (area of circular arc)

$\Rightarrow (400 - 100\pi) \text{ sq. m}$

Ans- (4) $(400 - 100\pi) \text{ sq. m.}$

14. $4 - \sin^2 45^\circ = 3.5$

$\cot K \tan 60^\circ$

$\Rightarrow 4 - \left(\frac{1}{\sqrt{2}}\right)^2 = 3.5$

$\frac{\cot K \sqrt{3}}{\sqrt{3}}$

$\Rightarrow 4 - \frac{1}{2} = 3.5 (\cot K \sqrt{3})$

$\Rightarrow \frac{8-1}{2} = 3.5 (\cot K \sqrt{3})$

$$\Rightarrow \frac{7}{2 \times 2.5} = (\cot k \sqrt{3})$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \cot k$$

$$\therefore k = 60^\circ$$

Ans- (3) 60°

15. $n = 57$

$$\frac{n}{2} = \frac{57}{2} = 28.5$$

Median class - 11.5 - 17.5

Ans- (2) 17.5

16. $4(\sin^4 30^\circ + \cos^4 60^\circ) - 3(\cos^2 45^\circ - \tan^2 45^\circ)$

$$\Rightarrow 4\left(\left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4\right) - 3\left(\left(\frac{1}{\sqrt{2}}\right)^2 - (1)^2\right)$$

$$\Rightarrow 4\left(\frac{1}{16} + \frac{1}{16}\right) - 3\left(\frac{1}{2} - 1\right)$$

$$\Rightarrow 4 \times \frac{1}{8} + \frac{3}{2}$$

$$\Rightarrow \frac{4}{2} = 2$$

Ans- (3) 2

17. Mode = 3Median - 2Mean

$$\Rightarrow 45 = 3\text{Median} - (2 \times 27)$$

$$\Rightarrow 45 + 54 = 3\text{Median}$$

$$\Rightarrow 33 = \text{Median}$$

Ans- (4) 33

18. $\frac{4}{3} \pi r^3 = 12\pi$

$\Rightarrow r^3 = \frac{12\pi \times 3}{4\pi} = 9$

$\Rightarrow r = \sqrt[3]{9} = 3^{2/3} \text{ cm}$

Ans- (3) $3^{2/3} \text{ cm}$

19. (2) Both Assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

20. (2) Both Assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A)

21. For real and equal roots, $b^2 - 4ac = 0$

$\Rightarrow x^2 + 2xk + k^2 - k + 2 = 0$

$\Rightarrow b^2 - 4ac = (2k)^2 - 4(k^2 - k + 2) = 0$

$\Rightarrow 4k^2 - 4k^2 + 4k + 2 = 0$

$\Rightarrow k = \frac{-2}{4} = -\frac{1}{2}$

22. In $\triangle ABD$, since $AB \parallel PQ$

By BPT, $\frac{z}{x} = \frac{QD}{BD}$

Similarly in $\triangle BDC$, $\frac{z}{y} = \frac{BQ}{BD}$

$\Rightarrow \frac{z}{x} + \frac{z}{y} = \frac{QD}{BD} + \frac{BQ}{BD}$

$\Rightarrow 2(\frac{1}{x} + \frac{1}{y}) = \frac{BD}{BD} = 1$ ($\because (QD + BQ) = BD$)

$\Rightarrow (\frac{1}{x} + \frac{1}{y}) = \frac{1}{2}$ Hence, proved

study time

Subject

$$23. \sin \theta = \frac{a}{b} = \frac{P}{H}$$

$$\Rightarrow P = aK, H = bK$$

$$B = \sqrt{H^2 - P^2}$$

$$\Rightarrow B = \sqrt{b^2 K^2 - a^2 K^2}$$

$$\Rightarrow B = (\sqrt{b^2 - a^2}) K$$

$$\tan \theta = \frac{P}{B}$$

$$\Rightarrow \frac{aK}{(\sqrt{b^2 - a^2})K} = \frac{a}{\sqrt{b^2 - a^2}}$$

24. Let inner radius be r and outer radius be R .

$$\text{Then, } 2\pi r = 352$$

$$\Rightarrow r = \frac{352 \times 7}{2 \times 22} = 56 \text{ cm}$$

$$2\pi R = 396$$

$$\Rightarrow R = \frac{396 \times 7}{2 \times 22} = 63 \text{ cm}$$

$$\text{Width of track} = R - r = 63 - 56 = \underline{\underline{7 \text{ cm}}}$$